

Module 3 – Operations Research

Question Bank

2 Marks Questions (Short / Theory)

1. Define Transportation Problem (TP).
2. Define Assignment Problem (AP).
3. Differentiate between Transportation Problem and Assignment Problem.
4. What is degeneracy in TP or AP?
5. What do we mean by balanced and unbalanced assignment problems?
6. What is Northwest Corner Method?
7. State the main objective of a TP.
8. Define Hungarian Method.
9. Difference between maximization and minimization in AP.
10. Give one real-life example of TP and one of AP.

4 Marks Questions (Moderate / Short Numericals)

1. Solve the LPP: Maximize $Z = 3x + 5y$, subject to: $x + y \leq 4$, $x \leq 2$, $y \leq 3$, $x, y \geq 0$.
2. Find an initial basic feasible solution of TP using Northwest Corner Method:
Warehouses W1,W2,W3 with supplies 30,50,20; Stores S1,S2,S3 with demands 40,30,30. Costs: W1[5,8,6], W2[4,7,3], W3[6,4,5].
3. Find an initial feasible solution using Least Cost Method for the same TP above.
4. Explain with a small example what is degeneracy in TP.
5. Solve 3x3 AP using Hungarian Method:
Workers W1-W3, Jobs J1-J3, Costs: W1[2,3,1], W2[4,2,3], W3[3,1,2].
6. What is Vogel's Approximation Method (VAM)? Explain penalty calculation with a small example.

8 Marks Questions (Long / Numerical + Theory)

1. Transportation Problem – Full Numerical: Warehouses W1,W2,W3 with supplies 20,30,25; Stores S1,S2,S3 with demands 30,25,20; Costs: W1[8,6,10], W2[9,12,13], W3[14,9,16]. Formulate TP, solve using VAM, compute total cost, check for degeneracy.
2. Assignment Problem – Full Numerical: Workers W1-W4, Jobs J1-J4, Costs: W1[9,2,7,8], W2[6,4,3,7], W3[5,8,1,8], W4[7,6,9,4]. Solve using Hungarian Method to minimize total cost, convert to maximization, explain difference.
3. Explain differences between TP and AP w.r.t. Objective, Decision variables, Supply & demand constraints, Real-life examples.
4. Unbalanced Assignment Problem: Workers W1-W3, Jobs J1-J5, Costs: W1[9,2,7,8,6], W2[6,4,3,7,5], W3[5,8,1,8,6]. Convert to balanced problem, solve using Hungarian Method, find minimum total cost.
5. Explain concept of penalty in VAM and why it leads to better initial solutions than Northwest Corner Method.